

Parcial I (Solución A)

(1)

1) a) Sea $X_1, \dots, X_n$ t.q. $X_i \sim \text{Bern}(p)$, i.e.

$$p(x_i) = p^{x_i} (1-p)^{1-x_i}, \quad x_i = 0, 1, \quad i = 1, \dots, n.$$

$$\Rightarrow \ln(p(x)) = x \ln p + (1-x) \ln(1-p)$$

$$\begin{aligned} \frac{\partial \ln(p(x))}{\partial p} &= \frac{x}{p} - \frac{1-x}{1-p} = \frac{(1-p)x - p(1-x)}{p(1-p)} \\ &= \frac{x - p}{p(1-p)} \end{aligned}$$

$$\begin{aligned} \Rightarrow E\left[\left(\frac{\partial \ln(p(x))}{\partial p}\right)^2\right] &= E\left[\frac{x^2 - 2xp + p^2}{p^2(1-p)}\right] \\ &= \frac{1}{p^2(1-p)^2} \left\{ E(x^2) - 2p E(x) + p^2 \right\} \end{aligned}$$

Como $\text{Var}(X) = pq = E(x^2) - p^2$, tenemos que

$$E(x^2) = pq + p^2 = p[(1-p) + p] = p.$$

$$\Rightarrow E\left[\left(\frac{\partial \ln(p(x))}{\partial p}\right)^2\right] = \frac{1}{p^2(1-p)^2} \left\{ p - 2p^2 + p^2 \right\} = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}$$

$$\begin{aligned} \Rightarrow \text{Var}[\hat{p}] &= \frac{p(1-p)}{n} = \frac{1}{n E\left[\left(\frac{\partial \ln(p(x))}{\partial p}\right)^2\right]} \\ &= \frac{\text{Var}(X)}{n}. \end{aligned}$$

$$\Rightarrow \hat{p} \text{ es EIVM de } p. //$$

$$b) \hat{p} \text{ es consistente pues } \text{Var}[\hat{p}] \xrightarrow{n \rightarrow \infty} 0 //$$

2) a) Se sabe que $X \sim \text{Potencial}(2, \theta)$

$$\Rightarrow E(X) = \frac{2\theta}{2+1} = \frac{2\theta}{3}.$$

El E.M es dado por la solución de.

$$\mu_1' = \frac{2}{3}\theta = \bar{X}$$

$$\Rightarrow \hat{\theta}_{EM} = \frac{3}{2} \bar{X} //$$

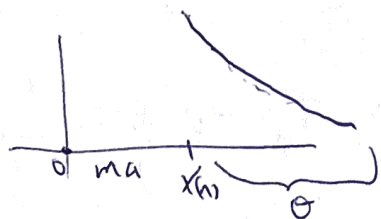
$$b) L(\theta) = \frac{2^n \prod_{i=1}^n x_i}{\theta^{2n}}.$$

$$\Rightarrow l(\theta) = n \ln 2 + \ln\left(\prod_{i=1}^n x_i\right) - 2n \ln(\theta)$$

$$\Rightarrow \frac{\partial l(\theta)}{\partial \theta} = -\frac{2n}{\theta} < 0.$$

$\Rightarrow$ no tiene punto crítico. y es siempre ↘.

Analizando el dominio de la función



el máximo es el menor θ posible pero no puede ser menor que $x_{(n)}$, entonces el máximo de $l(\theta)$ alcanza

cuando $\hat{\theta}_{MV} = x_{(n)}$.

$\therefore \hat{\theta}_{MV} = x_{(n)}$ es el EMV de θ .

c) Sabemos que $X_{(n)} \sim \text{Potencial}(2n, \theta)$.

L3

$$\Rightarrow E[X_{(n)}] = \frac{2n\theta}{2n+1} \quad \text{y} \quad B(X_{(n)}) = \frac{2n\theta}{2n+1} - \theta = \frac{2n\theta - 2n\theta - \theta}{2n+1} = \frac{-\theta}{2n+1}$$

y $X_{(n)}$ es insesgado.

$$\text{Adem\'as, } \text{Var}[X_{(n)}] = \frac{2n\theta^2}{(2n+1)^2(2n+2)}.$$

Por otro lado, el E.M de θ es $\hat{\theta}_{EM} = \frac{3}{2}\bar{X}$. Tenemos

que $E\left[\frac{3}{2}\bar{X}\right] = \frac{3}{2} \cdot \frac{2}{3}\theta = \theta$. es insesgado.

$$\text{y } \text{Var}\left[\frac{3}{2}\bar{X}\right] = \frac{9}{4} \frac{\text{Var}[X]}{n} = \frac{9}{4n} \frac{2\theta^2}{3^2 \cdot 4} = \frac{2\theta^2}{16n}.$$

El error cuadr\'atico medio de $X_{(n)}$ es:

$$\begin{aligned} ECM(X_{(n)}) &= \text{Var}(X_{(n)}) + B(X_{(n)})^2 \\ &= \frac{2n\theta^2}{(2n+1)^2(2n+2)} + \frac{\theta^2}{(2n+1)^2} = \frac{2\theta^2}{(2n+1)(2n+2)} \end{aligned}$$

y el ECM de $\hat{\theta}_{EM} = \frac{3}{2}\bar{X}$ es.

$$ECM(\hat{\theta}_{EM}) = \text{Var}[\hat{\theta}_{EM}] = \frac{2\theta^2}{16n}.$$

Si $n=2$

$$ECM(X_{(2)}) = \frac{2\theta^2}{(4+1)6} = \frac{\theta^2}{15} \quad \text{y} \quad ECM(\hat{\theta}_{EM}) = \frac{2\theta^2}{16 \cdot 2} = \frac{\theta^2}{16}.$$

$$\Rightarrow ECM(X_{(2)}) > ECM(\hat{\theta}_{EM})$$

Pero a partir de $n=3$,

$$ECM(X_{(n)}) < ECM(\hat{\theta}_{EM})$$

pues $ECM(X_{(n)})$ tiene grado 2 en el denominador y

$ECM(\hat{\theta}_{EM})$ tiene .. 1

3) a) Tenemos que $X_{(1)} \sim \text{Pareto}(\alpha n, \beta)$.

$$\text{i.e. } f_{X_{(1)}}(x) = \begin{cases} \frac{n\alpha \beta^{\alpha n}}{x^{n\alpha+1}} & , x \geq \beta \\ 0 & x < \beta \end{cases}$$

$$\text{Tome } U = \frac{X_{(1)}}{\beta} \Rightarrow h^{-1}(u) = \beta u \quad \text{y} \quad \frac{\partial h^{-1}(u)}{\partial u} = \beta.$$

$$\begin{aligned} \Rightarrow f_U(u) &= f(h^{-1}(u)) \left| \frac{\partial h^{-1}(u)}{\partial u} \right| \\ &= \frac{n\alpha \beta^{\alpha n}}{(\beta u)^{n\alpha+1}} |\beta| = \frac{n\alpha}{u^{n\alpha+1}} \quad , u \geq 1. \end{aligned}$$

$$\Rightarrow U = \frac{X_{(1)}}{\beta} \sim \text{Pareto}(\alpha n, 1)$$

satisface las condiciones de un pivote. pues su dist. no depende de β .

$$F_U(u) = \begin{cases} 1 - \left(\frac{1}{u}\right)^{\alpha n} & u \geq 1 \\ 0 & u < 1 \end{cases}$$

$$b) P(U \leq b) = 1 - \lambda.$$

$$\Rightarrow P\left(\frac{X_{(1)}}{\beta} \leq b\right) = 1 - \lambda$$

$$\Rightarrow 1 - \left(\frac{1}{b}\right)^{\alpha n} = 1 - \lambda$$

$$\Rightarrow \left(\frac{1}{b}\right)^{\alpha n} = \lambda \Rightarrow \left(\frac{1}{\lambda}\right)^{\frac{1}{\alpha n}} = b.$$

$$\Rightarrow P\left(\frac{X_{(1)}}{\beta} \leq \left(\frac{1}{\lambda}\right)^{\frac{1}{\alpha n}}\right) = 1 - \lambda.$$

despejando β , tenemos que

$$P\left(\frac{X_{(1)}}{\left(\frac{1}{\lambda}\right)^{\frac{1}{\alpha n}}} \leq \beta\right) = 1 - \lambda \Rightarrow \left(\frac{X_{(1)}}{\left(\frac{1}{\lambda}\right)^{\frac{1}{\alpha n}}}, +\infty\right) \text{ es un IC unilateral inferior de } (1-\lambda)\% \text{ para } \beta.$$